10.6 Asymptotic States in the LSZ Formula

The derivation of the LSZ formula in LecNote4 treats $a_1^\dagger(\pm\infty)$ as creating one normalized particle, $a_1(\pm\infty)$ as annihilating the vacuum, and the vacuum as the same state at both ends. This note works out what the limit $t \to \pm\infty$ does and does not do. Two of the conditions on the field, $\langle 0|\varphi|0\rangle = 0$ and $\langle p|\varphi(0)|0\rangle = 1$, have to be imposed by hand, while the multi-particle condition takes care of itself; the reason is one formula, Eq. (10.6.2). We then estimate how fast the multi-particle part goes away, say what $\langle 0|0\rangle = 1$ means for the in and out vacua, and identify what the time-ordering step on LecNote4 p. 5 drops.

Notation follows LecNote4 and Srednicki §5: metric $(-,+,+,+)$, $kx = \mathbf{k}\cdot\mathbf{x} - \omega t$, $\omega(\mathbf{k}) = \sqrt{\mathbf{k}^2 + m^2}$, $\widetilde{dk} = d^3k/\big((2\pi)^3 2\omega\big)$, and

$$a_1^\dagger(t) = \int d^3k\, f_1(\mathbf{k})\, a^\dagger(\mathbf{k},t), \qquad a^\dagger(\mathbf{k},t) = -i\int d^3x\, e^{ikx} \overleftrightarrow{\partial_0}\, \varphi(x), \tag{10.6.1}$$

with $f_1(\mathbf{k}) \propto \exp[-(\mathbf{k}-\mathbf{k}_1)^2/4\sigma^2]$, normalized so that in the free theory $\langle 0|a_1 a_1^\dagger|0\rangle = 1$.

One formula for every sector

Let $|s\rangle$ be any eigenstate of P^μ, with momentum $\mathbf{p}_s$ and energy E_s. Translation invariance gives $\langle s|\varphi(x)|0\rangle = e^{-ip_s x}\langle s|\varphi(0)|0\rangle$ (LecNote4 p. 7). Since $e^{ikx}\overleftrightarrow{\partial_0} e^{-ip_s x} = i(\omega + E_s)\, e^{i(k-p_s)x}$, the x integral in (10.6.1) gives $(2\pi)^3\delta^3(\mathbf{k}-\mathbf{p}_s)$ and

$$\langle s|a_1^\dagger(t)|0\rangle = (2\pi)^3\left[\omega(\mathbf{p}_s) + E_s\right] f_1(\mathbf{p}_s)\,\langle s|\varphi(0)|0\rangle\, e^{iK_s t}, \qquad K_s \equiv E_s - \omega(\mathbf{p}_s). \tag{10.6.2}$$

Integrated against a probe packet $\psi_n^*(\mathbf{p})$, this is (5.23) of LecNote4. K_s depends only on the eigenvalues of P^μ, so (10.6.2) is a matrix element of the operator statement

$$a_1^\dagger(t)|0\rangle = e^{iKt}\, a_1^\dagger(0)|0\rangle, \qquad K \equiv H - \sqrt{\mathbf{P}^2 + m^2}. \tag{10.6.3}$$

The time dependence of $a_1^\dagger(t)|0\rangle$ is generated by K, not by H. Two consequences follow.

First, K is Hermitian, so e^{iKt} is unitary and the norm of $a_1^\dagger(t)|0\rangle$ is the same at every t. Nothing is lost as $t \to \pm\infty$. The limit can only be a statement about overlaps with fixed states: a weak limit.

Second, for a fixed normalizable probe $|\psi\rangle$, the amplitude $\langle\psi|a_1^\dagger(t)|0\rangle$ is an average of e^{iKt} over the states in $|\psi\rangle$ and $a_1^\dagger(0)|0\rangle$. If K takes a continuum of values over those states, the phases cancel and the Riemann–Lebesgue

lemma (§10.5) sends the average to zero. If K takes a single value, the average is that one phase times a constant and does not decay. So the limit $t \to \pm\infty$ removes the continuous spectrum of K (strictly, its absolutely continuous part, which is all that smooth packets see) and nothing else.

Sector by sector

Vacuum. $E = 0$ and $\mathbf{p} = 0$, so $K = -m$. It is a single state, and there is nothing to average over. With $v \equiv \langle 0|\varphi(0)|0\rangle$, (10.6.2) gives

$$\langle 0|a_1^\dagger(t)|0\rangle = (2\pi)^3\, m\, v\, f_1(\mathbf{0})\, e^{-imt}. \tag{10.6.4}$$

The modulus is constant, and there is no limit unless $v f_1(\mathbf{0}) = 0$. For a generic packet this means $v = 0$, imposed by the shift $\varphi \to \varphi - v$ (LecNote4 p. 7).

One particle. $E = \omega(\mathbf{p})$ exactly, because $a^\dagger$ was built with the same m. So $K = 0$ for every momentum on the mass shell: the one-particle states form a continuum of H but a single point, $K = 0$, of K. For a one-particle probe $\langle\psi| = \int d^3p\, \psi^*(\mathbf{p})\langle p|$,

$$\langle\psi|a_1^\dagger(t)|0\rangle = (2\pi)^3 \int d^3p\, \psi^*(\mathbf{p})\, 2\omega\, f_1(\mathbf{p})\, \langle p|\varphi(0)|0\rangle, \tag{10.6.5}$$

independent of t. By Lorentz invariance $\langle p|\varphi(0)|0\rangle$ is a constant (LecNote4 p. 7). The one-particle part of $a_1^\dagger(t)|0\rangle$ is this constant times the free-theory packet, at every t, and the limit cannot change it. Its normalization is fixed by the rescaling $\langle p|\varphi(0)|0\rangle = 1$ (LecNote4 p. 8).

Two or more particles. A state $|p, n\rangle$ with total momentum $\mathbf{p}$ has $E = \sqrt{\mathbf{p}^2 + M^2}$, with invariant mass $M \geq 2m$ running over a continuum. Then $K = \sqrt{\mathbf{p}^2 + M^2} - \sqrt{\mathbf{p}^2 + m^2} > 0$ varies with $\mathbf{p}$ and with M, and (5.23) goes to zero. No condition on φ is needed. The same holds for a bound state of mass $M \neq m$: it is an isolated point of the mass spectrum, but its K still varies with $\mathbf{p}$.

The physical picture is the one on LecNote4 p. 10. The operator $a_1^\dagger(t)$ builds its packet with the one-particle dispersion $\omega(\mathbf{k})$. The part of $a_1^\dagger(t)|0\rangle$ that really is one particle moves with that dispersion and stays with the packet. The multi-particle part moves with group velocity $\mathbf{p}/\sqrt{\mathbf{p}^2 + M^2} \neq \mathbf{p}/\omega$ and drifts away from any fixed state. The vacuum cannot drift; it is everywhere.

Annihilation operator. The same algebra with $a(\mathbf{k}, t) = i \int d^3x\, e^{-ikx} \overleftrightarrow{\partial_0} \varphi(x)$ gives

$$\langle s|a_1(t)|0\rangle = (2\pi)^3 \left[\omega(\mathbf{p}_s) - E_s\right] f_1^*(-\mathbf{p}_s)\, \langle s|\varphi(0)|0\rangle\, e^{i(\omega(\mathbf{p}_s) + E_s)t}. \tag{10.6.6}$$

The one-particle part vanishes identically, at every t, because $E_s = \omega$. The vacuum part is $\propto v\, e^{imt}$ and needs $v = 0$. The multi-particle part dephases. Putting the sectors together, with $v = 0$ and $\langle p|\varphi(0)|0\rangle = 1$,

$$a_1^\dagger(t)|0\rangle \;\to\; \int d^3k\, f_1(\mathbf{k})\,|k\rangle, \qquad a_1(t)|0\rangle \;\to\; 0 \qquad (t \to \pm\infty, \text{ weakly}). \quad (10.6.7)$$

These are the properties the derivation in LecNote4 uses. The limit is weak: the multi-particle part of $a_1^\dagger(t)|0\rangle$ keeps its norm, it just stops overlapping with anything fixed. For one operator acting on the vacuum this is all LSZ needs. Making it rigorous for products of several $a^\dagger(-\infty)$ is the content of Haag–Ruelle scattering theory. (Strictly, in an interacting theory in four dimensions a field smeared in space at a single time is too singular for the norm of $a_1^\dagger(t)|0\rangle$ to be finite; Haag–Ruelle theory smears in time as well. The conclusions carry over.)

The same projection in momentum space

LecNote11 (pp. 5–7) brings the multi-particle states back as the continuum $\rho(s)$ of the exact propagator $\tilde{\mathbf{\Delta}}$. "Didn't the limit remove them?"

They never left the field. As §10.15 ("LSZ and the continuum") shows, $|\langle k, n|\varphi(x)|0\rangle|$ is the same at every x; what dephases, the superposition of the phases $e^{iK_s t}$ of (10.6.2), needs a packet built with the one-particle $\omega(\mathbf{k})$ and $t \to \pm\infty$, and LecNote11 p. 5 has neither (its red box: "Here is not that limit"). In momentum space LSZ's factor $k^2 + m^2$ removes the continuum of the Lehmann–Källén form at the pole, Eq. (10.15.6), and leaves the residue $|\langle k|\varphi(0)|0\rangle|^2 = 1$, the factor 1 per external line of §10.11. At fixed spatial momentum $\mathbf{p}$, with $E = \sqrt{\mathbf{p}^2 + M^2}$ and $\omega = \omega(\mathbf{p})$, LSZ's factor turns each term $1/(k^2 + s - i\epsilon)$, $s = M^2$, into

$$\frac{k^2 + m^2}{k^2 + M^2} = \frac{k^0 - \omega}{k^0 - E} \cdot \frac{k^0 + \omega}{k^0 + E}.$$

At LSZ's $k^0 = \omega$ this is 1 for $M = m$ and 0 otherwise. The pole at $k^0 = E$ misses the one-particle shell (the red box's "off-shell") by $E - \omega = K_s$. With $v = 0$, both pictures keep exactly $K = 0$: in time by dephasing, in momentum by the zero of $k^2 + m^2$. For $\rho(s)$ itself, see §10.15.

How fast the multi-particle part goes away

To see the rates, take the probe equal to the packet and treat the smooth prefactor in (10.6.2) as constant across it ($\sigma \ll m$). Normalized to $A(0) = 1$,

$$A(t) = \int d^3p\, w(\mathbf{p})\, e^{iK(\mathbf{p})t}, \qquad w(\mathbf{p}) = (2\pi\sigma^2)^{-3/2}\, e^{-(\mathbf{p}-\mathbf{k}_1)^2/2\sigma^2}, \quad (10.6.8)$$

with $K(\mathbf{p}) = E_s(\mathbf{p}) - \omega(\mathbf{p})$ for the sector in question; $w = |f_1|^2$ up to normalization. For a fixed-mass sector, $E_s = \sqrt{\mathbf{p}^2 + M^2}$.

Gaussian estimate. Expand K about the packet centre. The size of its gradient there is the difference of group velocities,

$$\Delta v = \frac{k_1}{\omega(k_1)} - \frac{k_1}{\sqrt{k_1^2 + M^2}}, \tag{10.6.9}$$

and keeping only the linear term turns (10.6.8) into the Fourier transform of a Gaussian:

$$|A(t)| \approx e^{-(\sigma\,\Delta v\, t)^2/2}, \qquad \tau = \frac{1}{\sigma\,\Delta v}. \tag{10.6.10}$$

The two components separate at relative speed Δv and stop overlapping once the separation exceeds the packet size, about $1/\sigma$. The two directions transverse to $\mathbf{k}_1$ add a factor $\left[1 + \left((1/\omega - 1/E_s)\sigma^2 t\right)^2\right]^{-1/2}$, which only speeds up the decay. The cubic term in the expansion corrects (10.6.10) once $|A|$ is down to a few per cent.

Power-law tail. The gradient of K vanishes at $\mathbf{p} = 0$, where both components are at rest and nothing moves apart. Near that point $K \approx (M - m) - \frac{1}{2}(1/m - 1/M)\,\mathbf{p}^2$. For a packet at rest ($\mathbf{k}_1 = 0$) the Gaussian integral is exact at this order:

$$A(t) = e^{i(M-m)t}\left[1 + i(1/m - 1/M)\,\sigma^2 t\right]^{-3/2}, \qquad |A| \;\to\; \left[(1/m - 1/M)\,\sigma^2 t\right]^{-3/2}. \tag{10.6.11}$$

For a moving packet, (10.6.10) dies first. The small part of the packet near $\mathbf{p} = 0$, weighted by $w(\mathbf{0}) \propto e^{-k_1^2/2\sigma^2}$, then takes over and falls only as a power of t. In a one-dimensional model along $\mathbf{k}_1$ the stationary-phase estimate of this tail is

$$|A_{\text{tail}}| \approx w(0)\left[\frac{2\pi}{(1/m - 1/M)\,t}\right]^{1/2}. \tag{10.6.12}$$

Continuum in M. At fixed $\mathbf{p}$ the multi-particle sector also has a continuum of M, so it dephases further. Near threshold the two-particle weight typically grows like $(M - 2m)^{1/2}$, which gives another $t^{-3/2}$.

Wrong mass. If the m used to build $a^\dagger(\mathbf{k})$ is not the pole mass m', the one-particle $K = \sqrt{\mathbf{p}^2 + m'^2} - \sqrt{\mathbf{p}^2 + m^2}$ varies with $\mathbf{p}$. The one-particle overlap then decays too, following (10.6.10) with $M \to m'$, and the LSZ formula returns zero. This is the dynamical reason the m in LSZ must be the physical mass (LecNote4 p. 10).

Numbers. The table on p. 44 and Fig. 10.1 evaluate (10.6.8) in the one-dimensional model along $\mathbf{k}_1$, with $m = 1$, $k_1 = m$ and $\sigma = 0.25\,m$. For the shell at $M = 2m$, $\Delta v = 1/\sqrt{2} - 1/\sqrt{5} = 0.260$ and $\tau = 15.4$; for a packet

mass 5% off ($m' = 1.05\,m$), $\Delta v = 0.0175$ and $\tau = 229$. The Gaussian estimate describes the 5% case throughout. For $M = 2m$ it holds to $t \approx 30$; the cubic term $K'''(k_1) = 0.316$ accounts for the gap at $t = 40$ (the quadratic term, $K'' = 0.004$, is negligible). From $t \approx 60$ on the power-law tail takes over. Eq. (10.6.12), with $w(0) = 5.4 \times 10^{-4}$, is its asymptotic form: it is within 30% of the exact value at $t = 160$ and within 10% for $t \gtrsim 300$, because w varies quickly across the region near $p = 0$. In three dimensions with the packet at rest, the numerical $|A|\,t^{3/2}$ approaches 180.9 at $t = 800$, against 181.0 from (10.6.11).

	packet mass 5% off		shell at $M = 2m$			
$m\,t$	exact	Eq. (10.6.10)	exact	Eq. (10.6.10)	cubic K	Eq. (10.6.12)
20	0.996	0.996	0.444	0.430	0.443	–
40	0.986	0.985	2.0×10^{-2}	3.4×10^{-2}	2.1×10^{-2}	–
80	0.944	0.941	4.8×10^{-4}	1.4×10^{-6}	–	2.1×10^{-4}
160	0.794	0.784	1.9×10^{-4}	3×10^{-24}	–	1.5×10^{-4}

$|A(t)|$ **IN THE ONE-DIMENSIONAL MODEL**

From (10.6.8) in one dimension, $m = 1$, $k_1 = m$, $\sigma = 0.25\,m$. The vacuum ($|A| = |e^{-imt}|$) and the one-particle sector with the correct mass ($K \equiv 0$) are exactly 1 at all t. The cubic expansion of K is shown only while it applies; at late times it has spurious stationary points far out in the packet.

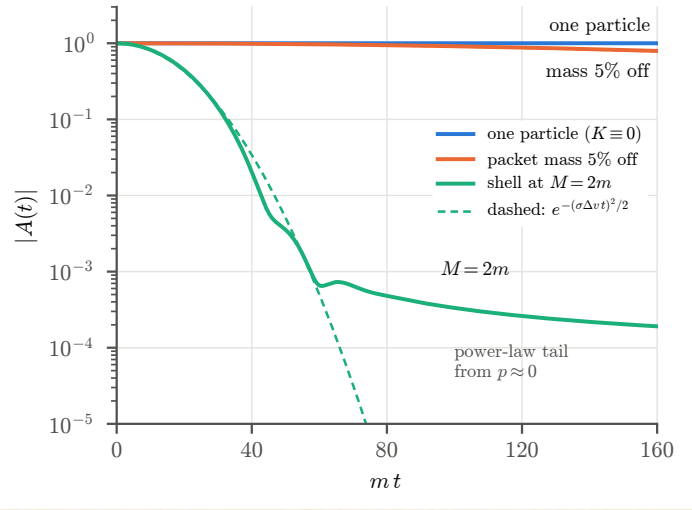


Figure 10.1: $|A(t)|$ for the one-particle sector ($K \equiv 0$), a packet built with a mass 5% off, and a shell at $M = 2m$; parameters as in the table on p. 44. Dashed: the Gaussian estimate (10.6.10) (for the 5% case it lies on the solid curve).

THE IN AND OUT VACUUM

On LecNote4 p. 5, after time ordering, the operators $a_{i'}(-\infty)$ act on the ket and the $a_j^\dagger(+\infty)$ act on the bra. So the ket must satisfy $a(-\infty)|0\rangle = 0$:

it is the in-vacuum. The bra must satisfy $\langle 0|a^\dagger(+\infty)=0$, i.e. $a(+\infty)|0\rangle=0$: it is the out-vacuum. In the Heisenberg picture states carry no time label; "in" and "out" say which asymptotic operators annihilate the state. By (10.6.7), the exact ground state, with $v=0$, is annihilated at both ends. So $|0_{\rm in}\rangle=|0_{\rm out}\rangle=|0\rangle$, and

$$\langle 0|0\rangle = 1 \quad\Longleftrightarrow\quad \langle 0_{\rm out}|0_{\rm in}\rangle = 1. \tag{10.6.13}$$

The content is that the vacuum is stable: no particles are produced from it. That needs a unique ground state and a time-independent H. In the Heisenberg picture, with one vacuum state, no phase can appear. In the path integral the same amplitude is computed as $\langle 0|e^{-iHT}|0\rangle = e^{-iE_0T}$, and there it needs the choice $E_0=0$ (LecNote5 p. 8 and §10.8).

LecNote4 p. 7 uses that both vacua are annihilated by P^μ. The step $\langle 0|\varphi(x)|0\rangle=\langle 0|\varphi(0)|0\rangle$ uses $\langle 0|P^\mu=0$ on the bra and $P^\mu|0\rangle=0$ on the ket. That is why v is a constant, and why $v=0$ as $t\to\pm\infty$ is the same as $v=0$ everywhere.

A time-dependent external source shows what happens otherwise. Add $J(x)\varphi(x)$ to the free Lagrangian, with J switched on and off. The vacuum-to-vacuum amplitude is (Srednicki §8)

$$\langle 0_{\rm out}|0_{\rm in}\rangle_J = \exp\Big[\frac{i}{2}\int d^4x\, d^4y\, J(x)\Delta(x-y)J(y)\Big], \tag{10.6.14}$$

and with $\mathrm{Im}(k^2+m^2-i\epsilon)^{-1}=\pi\delta(k^2+m^2)$ its squared modulus is

$$\big|\langle 0_{\rm out}|0_{\rm in}\rangle_J\big|^2 = \exp\Big[-\int \widetilde{dk}\,\big|\tilde J(\omega,\mathbf{k})\big|^2\Big], \qquad \tilde J(k)=\int d^4x\, e^{-ikx}J(x). \tag{10.6.15}$$

This is less than 1 whenever J has Fourier components on the mass shell: the source creates particles, on average $\int\widetilde{dk}\,|\tilde J|^2$ of them. The in–out expectation value of the field,

$$\frac{\langle 0_{\rm out}|\varphi(x)|0_{\rm in}\rangle_J}{\langle 0_{\rm out}|0_{\rm in}\rangle_J} = \int d^4y\,\Delta(x-y)\,J(y), \tag{10.6.16}$$

depends on x: it solves $(-\partial^2+m^2)\varphi=J$. With time translation broken, the vacua are no longer eigenstates of P^0, the in and out vacua differ, and a condition "at $t\to\pm\infty$" is no longer the same as one "at all times". For $J=0$ both statements reduce to (10.6.13).

Time ordering and the overlap term

After substituting (5.11) and (5.12), LecNote4 p. 5 uses T to move every $a^\dagger(+\infty)$ to the far left and every $a(-\infty)$ to the far right, where they vanish by (10.6.7). What survives is the LSZ integral. It reproduces $\langle f|i\rangle$ up to

terms of order $e^{-(\mathbf{k}_{i'}-\mathbf{k}_j)^2/8\sigma^2}$: the step drops the part of $\langle f|i\rangle$ in which nothing scatters. Write the overlap of two of the one-particle packets in (10.6.7) as

$$O_{i'j} \equiv (2\pi)^3 \int d^3k\; 2\omega\; f^*_{i'}(\mathbf{k}) f_j(\mathbf{k}) \;\approx\; e^{-(\mathbf{k}_{i'}-\mathbf{k}_j)^2/8\sigma^2}, \tag{10.6.17}$$

where the Gaussian form follows from completing the square, with ω roughly constant across the packets. For $2 \to 2$ with stable particles,

$$\langle f|i\rangle = O_{1'1}O_{2'2} + O_{1'2}O_{2'1} + \big(\text{LSZ formula}\big). \tag{10.6.18}$$

The overlap terms are the part of $S = 1 + iT$ in which nothing scatters. Choosing every $|\mathbf{k}_{i'} - \mathbf{k}_j| \gg \sigma$, as on LecNote4 p. 3, makes them exponentially small. In the plane-wave limit they become the $\delta^3(\mathbf{k}_{i'} - \mathbf{k}_j)$ terms of the "1".

Free-theory check. In the free theory $(-\partial^2 + m^2)\varphi = 0$ as an operator, so (5.11) and (5.12) say $a^\dagger(-\infty) = a^\dagger(+\infty)$ and $a(+\infty) = a(-\infty)$. The integrals vanish, and the only term left after the substitution is $\langle 0|T\, a_{1'}(-\infty)a_{2'}(-\infty)a_2^\dagger(+\infty)a_1^\dagger(+\infty)|0\rangle$, which the step on p. 5 sets to zero. But these are the same four operators as in the original product, whose value is $O_{1'1}O_{2'2} + O_{1'2}O_{2'1}$. The labels $\pm\infty$ do not order them in any physical sense. The step drops exactly the no-scattering part, as (10.6.18) says.

In the interacting theory the dropped pieces come from the ends of the time integrals in (5.11) and (5.12), which run to the same asymptotic times as the operators they are ordered against. Individually these pieces also contain multi-particle terms; their total is the overlap terms in (10.6.18), which we do not show here. Every one of them contains $f^*_{i'}(\mathbf{p}) f_j(\mathbf{p})$ at a common momentum, so all of them vanish when every $|\mathbf{k}_{i'} - \mathbf{k}_j| \gg \sigma$.

A two-mode model. Take two oscillators of the same frequency with a coupling switched on and off, $H = \omega(a^\dagger a + b^\dagger b) + g(t)(a^\dagger b + b^\dagger a)$. In the rotating frame $\tilde a(t) = e^{i\omega t}a(t)$ the free evolution drops out, which is the job $(-\partial^2 + m^2)$ does in LSZ, and $\dot{\tilde a} = -ig\,\tilde b$, $\dot{\tilde b} = -ig\,\tilde a$. With $\theta(t) = \int_{-\infty}^{t} g\,dt'$ and $\Theta = \theta(\infty)$, $\tilde a(t) = \cos\theta\,\tilde a_{\rm in} - i\sin\theta\,\tilde b_{\rm in}$. The analogue of (5.12) is $\tilde a_{\rm out} = \tilde a_{\rm in} - i\int dt\, g(t)\,\tilde b(t)$, and

$$\langle 0|\tilde a_{\rm out}\tilde a_{\rm in}^\dagger|0\rangle = \underbrace{\langle 0|\tilde a_{\rm in}\tilde a_{\rm in}^\dagger|0\rangle}_{1} - \int dt\, g \sin\theta = \cos\Theta, \tag{10.6.19}$$

$$\langle 0|\tilde b_{\rm out}\tilde a_{\rm in}^\dagger|0\rangle = \underbrace{\langle 0|\tilde b_{\rm in}\tilde a_{\rm in}^\dagger|0\rangle}_{0} - i\int dt\, g \cos\theta = -i\sin\Theta. \tag{10.6.20}$$

The integral, the analogue of the LSZ integrand, gives the full amplitude when the initial and final states differ, and the amplitude minus the

overlap, $\cos\Theta - 1$, when they coincide. Moving $\tilde{a}_{\rm in}$ to the right, where it annihilates $|0\rangle$, is exactly the step that drops the underbraced 1.

Derivatives outside T. The factors $(-\partial^2 + m^2)$ in the LSZ formula act outside the time ordering. Moving ∂_0^2 inside produces contact terms from the equal-time commutator,

$$(-\partial_x^2 + m^2)\,\langle 0|T\varphi(x)\varphi(y)X|0\rangle \\ = \langle 0|T\big[(-\partial^2 + m^2)\varphi(x)\big]\varphi(y)X|0\rangle - \frac{i}{Z_\varphi}\,\delta^4(x-y)\,\langle 0|TX|0\rangle + \dots, \quad \textbf{(10.6.21)}$$

where X is the product of the remaining fields and the dots are the same contact term for each field in X. The factor Z_φ is the one on LecNote4 p. 10: for the field normalized by $\langle p|\varphi(0)|0\rangle = 1$ the canonical momentum is $Z_\varphi\partial_0\varphi$ (in the free theory $Z_\varphi = 1$). A contact term sets two positions equal, so its Fourier transform depends on the two external momenta only through one combination. It cannot have a separate pole in each of the two k^2, and for non-forward kinematics, $|\mathbf{k}_{i'} - \mathbf{k}_j| \gg \sigma$, the factors $k^2 + m^2 \to 0$ remove it on shell.